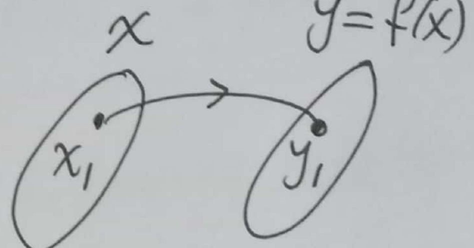
Calculus I Lecture #1

Definition of a mathematical Function

It's a special relation between two variables, say x and y, x is considered as an independent variable and y is the dependent variable. For each value of x, we've only one value for y.



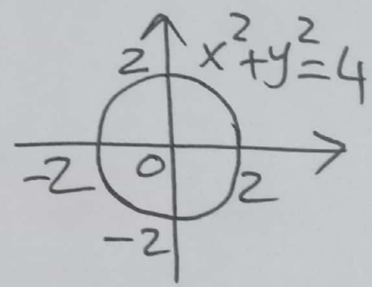
$\Rightarrow$ we write $y = f(x)$

Examples:

$y = x^3$ $\qquad$ $y = \sin x$ $\qquad$ $y = \sqrt{x}$

Remark:

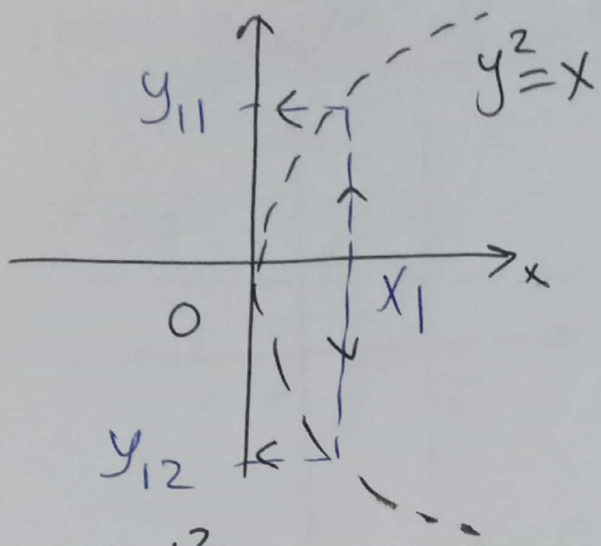
For the relation $x^2 + y^2 = 4$:



at $x = 0 \longrightarrow y = \pm 2$

So it's not a function because for $x = 0$ we've more than one value for y.

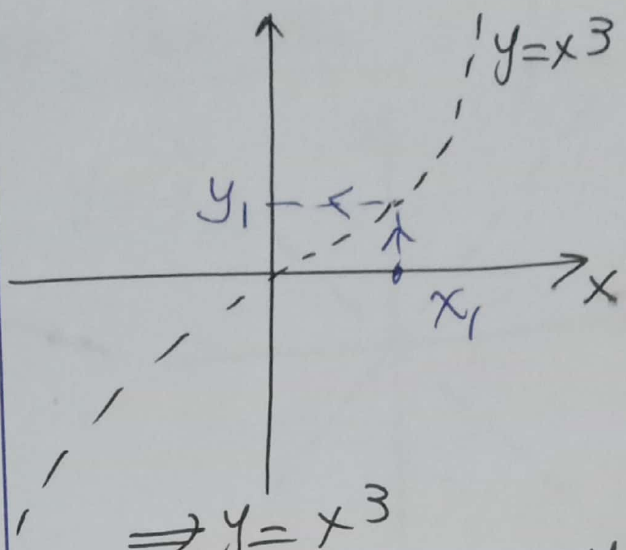
Also $\Rightarrow y^2 = x$ is not a function



$y^2 = x$

for $x_1 \Rightarrow$ More than one y_1

this not a function



$\Rightarrow y = x^3$

for $x_1 \longrightarrow$ only one y_1

This's a function

<u>Example</u>: which of the following relations can be considered as a function?

(i) $y^3 = x$ (ii) $y = \sin^{-1} x$ "shift sin"

<u>Solution</u>:

(i) for $x_1 \Rightarrow y_1^3 = x_1 \Rightarrow y_1 = \sqrt[3]{x_1}$ "one value"

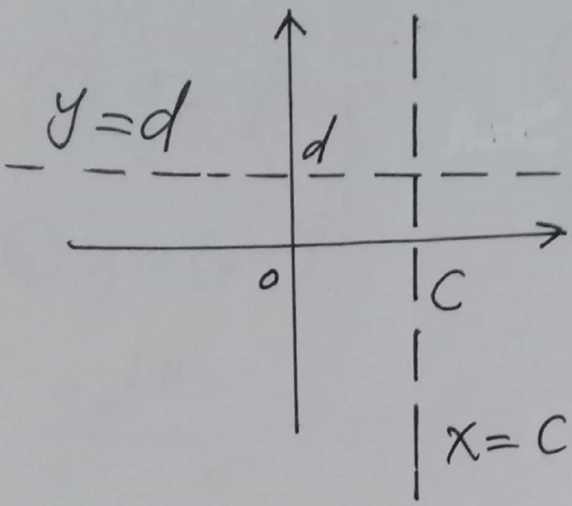
$\Rightarrow$ it's a function

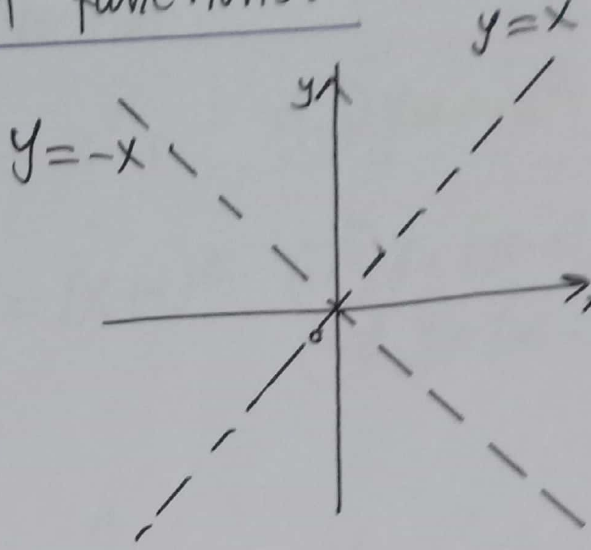
(ii) say $x_1 = 1 \Rightarrow y_1 = \sin^{-1} 1$

$\Rightarrow y_1 = \frac{\pi}{2}, \frac{5\pi}{2}, \frac{9\pi}{2} - - -$

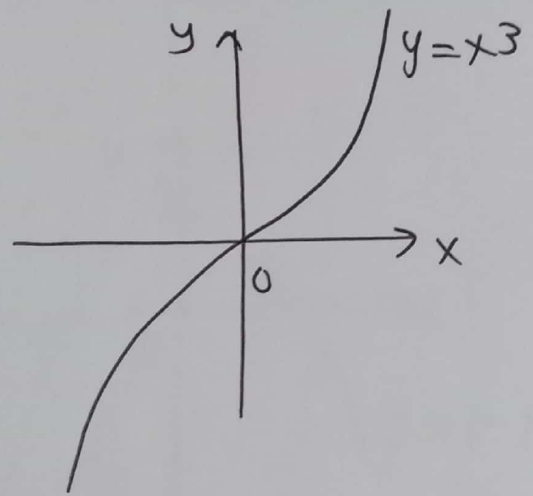
So it's <u>not</u> a function

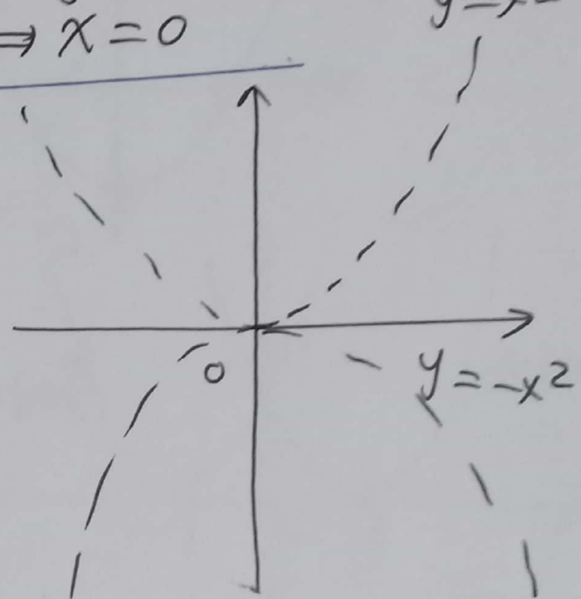
Some graphs of common functions:

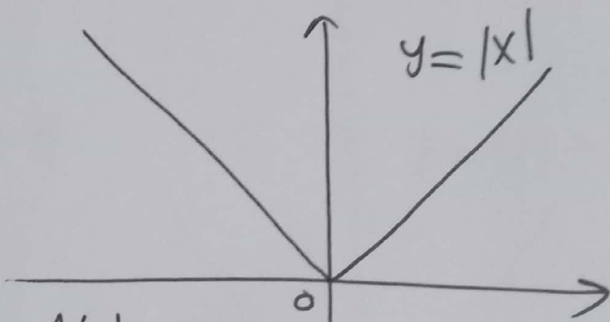




Note: Eq^n of x-axis $\Rightarrow y=0$
~ ~ y-axis $\Rightarrow x=0$







Note:

$$y=|x|=\begin{cases} x & x \geq 0 \\ -x & x<0 \end{cases}$$

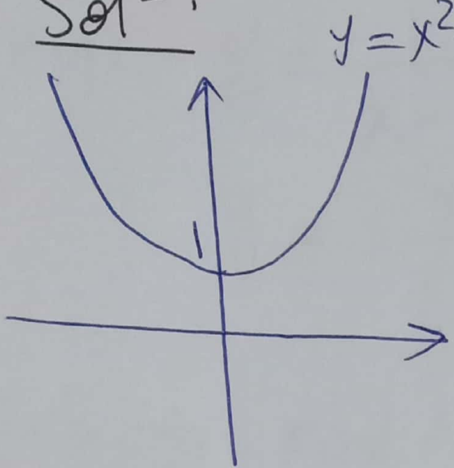
Example: Graph the function:

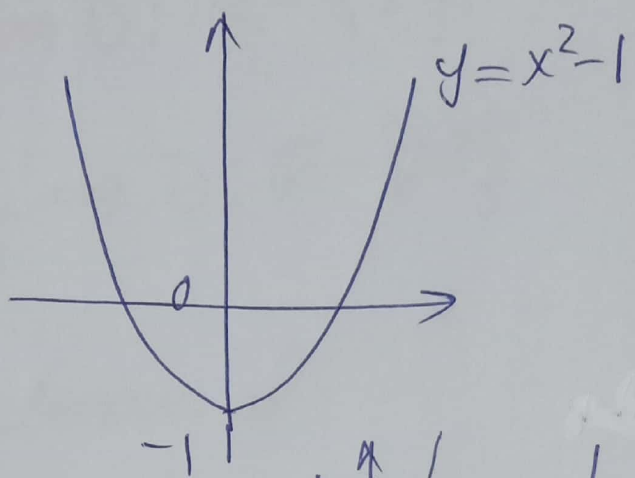
(i) $y = x^2 + 1$ (ii) $y = x^2 - 1$ (iii) $y = 1 - x^2$

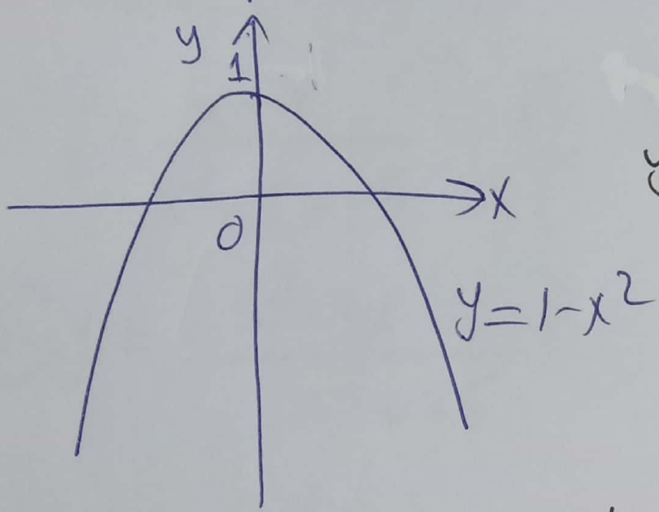
(v) $y = (x-1)^2$ (iv) $y = (x+1)^2$ (iiv) $y = |x-1|$

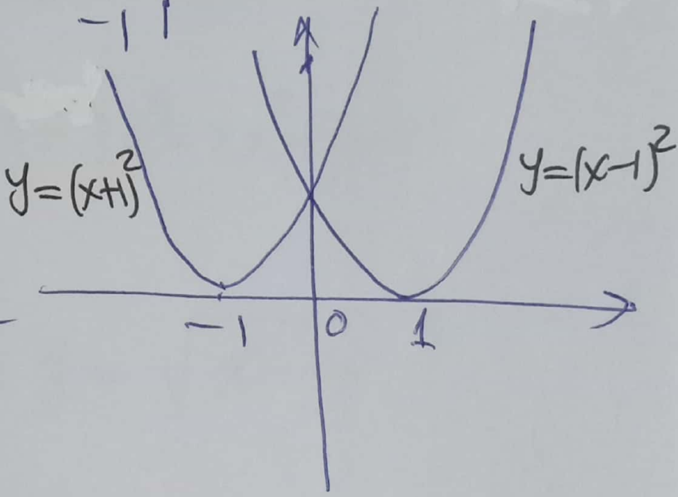
(iiv) $y = |x| - 1$

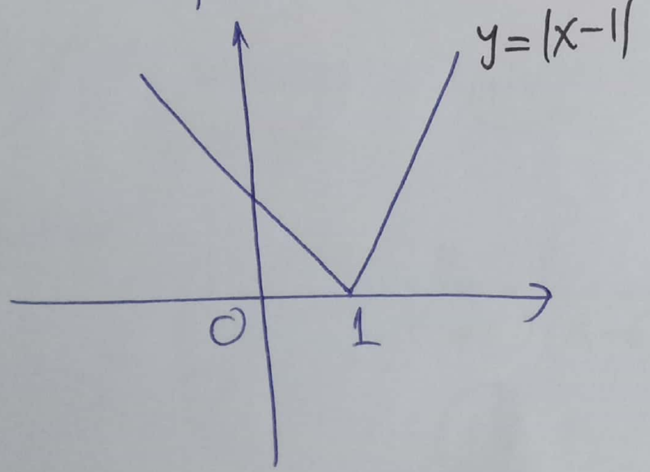
Sol^{n}:

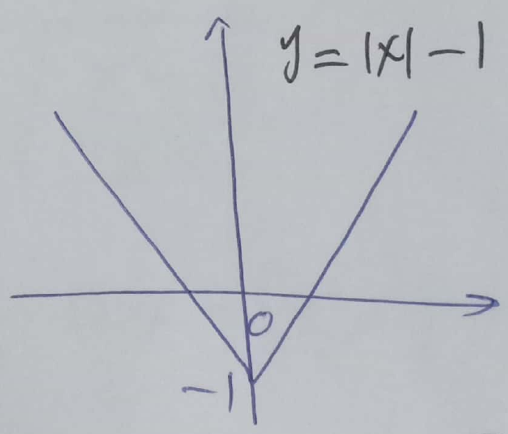












Domain of a function:

Let $y = f(x)$. The set of all values of x at which y is defined is called the domain of $f(x)$.

e.g. $y = \frac{1}{x-2} \Rightarrow D: R - \{2\}$

$y = \frac{x+1}{x^2-4} \Rightarrow D: R - \{\pm 2\}$

Examples: Find the domain of:

(i) $y = \frac{x+3}{x^2-5x-6}$

(ii) $y = 3x^3 - 2x^2 + 7x - 1$

(iii) $y = \frac{x^2-3x-4}{x^2+25}$

(iv) $y = \sqrt{x-4}$

(v) $y = \frac{x}{\sqrt{x+2}}$

(iv) $y = \sqrt{\frac{x-2}{x+1}}$

Solutions: (i) $y = \frac{x+3}{(x+1)(x-6)}$ $D: R - \{-1, 6\}$

(ii) $D: R$

(iii) $D: R$

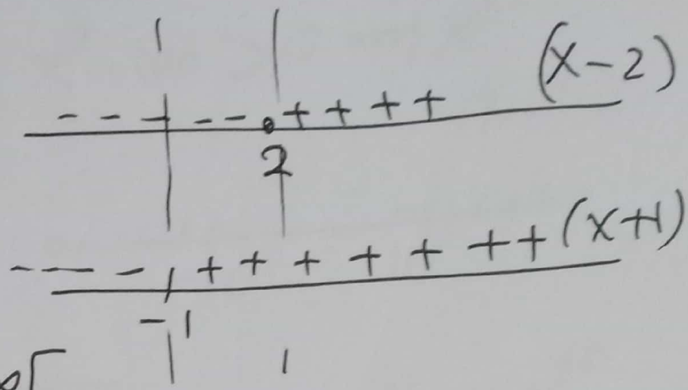
(iv) $y=\sqrt{x-4}$ D: $x-4\geq 0 \Rightarrow x\geq 4$

D: $[4,\infty[$

(v) $y=\dfrac{x}{\sqrt{x+2}}$ D: $x+2>0 \Rightarrow x>-2$

D: $]-2,\infty[$

(iiv) $y=\sqrt{\dfrac{x-2}{x+1}}$



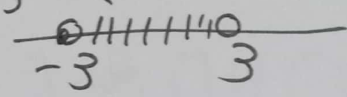
D: $]-\infty,-1[\cup[2,\infty[$

Remarks $\Rightarrow$ $x^2=9 \Rightarrow x=\pm 3$

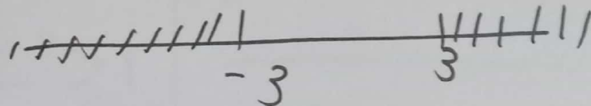
$|x|=3 \Rightarrow x=\pm 3$

then:

$x^2<9 \Rightarrow |x|<3 \Rightarrow -3<x<3$



$x^2>9 \Rightarrow |x|>3$



Ex. Find domain of.

(i) $y=\sqrt{16-x^2}$ (ii) $y=\dfrac{x^2-x+1}{\sqrt{x^2-100}}$

(iii) $y=\sqrt{x^2-4x+3}$

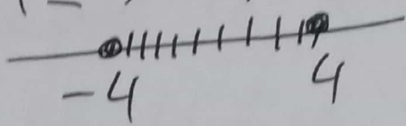
Solution:

(i) $y=\sqrt{16-x^2}$ D: $16-x^2 \geq 0$

$\Rightarrow x^2 \leq 16$

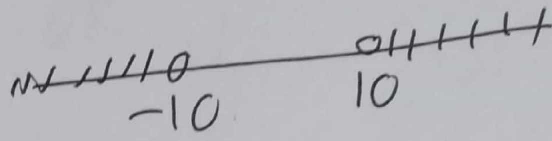
$\Rightarrow |x| \leq 4$

$-4 \leq x \leq 4$



(ii) $y=\dfrac{x^2-x+1}{\sqrt{x^2-100}}$ D: $x^2-100>0 \Rightarrow x^2>100$

$|x|>10$



(iii) $y=\sqrt{(x-2)^2-1}$

$(x-2)^2-1 \geq 0$

$(x-2)^2 \geq 1$

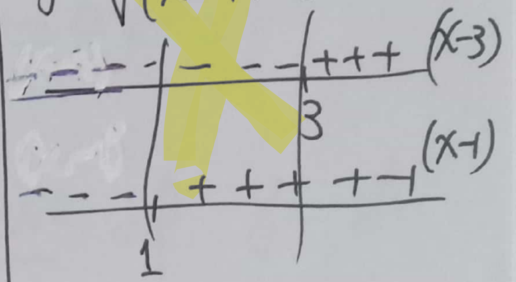
$|x-2| \geq 1$

$x-2 \leq -1$ $x-2 \geq 1$

$x \leq 1$ $x \geq 3$

Another Sol^n

$y=\sqrt{(x-1)(x-3)}$



$]-\infty, 1] \cup [3, \infty[$

Function Differentiation:

Let $y = f(x)$. The rate of change of y with respect to x can be obtained by differtiating y relative to $x \Longrightarrow \frac{dy}{dx}$ OR y'

e.g. $y = x^4 \Rightarrow y' = 4x^3$

we write $\frac{d}{dx} x^4 = 4x^3$

we say that the <u>derivative</u> of $y = x^4$ is $y' = 4x^3$

similarly, $y = x^{3/2} \Longrightarrow y' = \frac{3}{2} x^{\frac{1}{2}}$

OR $\frac{d}{dx} x^{3/2} = \frac{3}{2} x^{1/2}$

Also $\frac{d}{dx} x^{-5} = -5x^{-6}$

and so on

Rule 1 $\frac{d}{dx}x^n = nx^{n-1}$ OR $y = x^n$
$y' = nx^{n-1}$

Ex. $y = x^{10}$
$y' = 10x^9$

$y = \sqrt[3]{x} = x^{1/3}$
$y' = \frac{1}{3}x^{-2/3}$

$y = \frac{1}{x^2} = x^{-2}$
$y' = -2x^{-3}$

Rule 2: $\frac{d}{dx}c = 0$ $\frac{d}{dx}x = 1$

$\frac{d}{dx}cx^n = cnx^{n-1}$

Ex. $y = x$
$y' = 1$

$y = 4$
$y' = 0$

$y = 6x^8$
$y' = 6(8)x^7$
$= 48x^7$

$y = 4x^2$
$y' = 4(2)x$
$= 8x$

$y = 6x$
$y' = 6(1)$
$= 6$

$y = 1$
$y' = 0$

Property: $\frac{d}{dx}\left(f(x) \pm g(x)\right) = \frac{d}{dx} f(x) \pm \frac{d}{dx} g(x)$

Ex. $y = 6x^8 - 3x + 10$

$y' = 6(8)x^7 - 3(1) + 0 = 48x^7 - 3$

Ex. $y = 4\sqrt{x} - \frac{3}{x^3} + 9x - \frac{1}{2}$

$y' = 4\left(\frac{1}{2}\right)x^{-\frac{1}{2}} - 3(-3)x^{-4} + 9(1) - 0$

Ex. $y = \frac{1}{10}x^3 - \frac{1}{4x} + \frac{9}{\sqrt[4]{x}}$

$y' = \frac{1}{10}(3)x^2 - \frac{1}{4}(-1)x^{-2} + 9\left(-\frac{1}{4}\right)x^{-5/4}$

Rule: $\frac{d}{dx}(f(x))^n = n(f(x))^{n-1} f'(x)$

Ex. $y = (2x^8 - 6x + 7)^4$

$y' = 4(2x^8 - 6x + 7)^3 (2(8)x^7 - 6)$

Ex. $y = \frac{1}{(3 - 2x + x^3)^{10}} = (3 - 2x + x^3)^{-10}$

$y' = -10(3 - 2x + x^3)^{-11}(-2 + 3x^2)$

Ex. $y = \sqrt[3]{3x - 1} = (3x - 1)^{1/3}$

$y' = \frac{1}{3}(3x - 1)^{-2/3}(3)$

Ex. $y = \frac{1}{\sqrt[5]{2x^4 - 7x^2 + x}} = (2x^4 - 7x^2 + x)^{-\frac{1}{5}}$

$y' = \left(-\frac{1}{5}\right)(2x^4 - 7x^2 + x)^{-6/5}(8x^3 - 14x + 1)$

Rule:

$$\frac{d}{dx}\left[f(x).g(x)\right] = f'(x)\,g(x) + f(x)\,g'(x)$$

Ex. $y = (x^2+3x-1)^4(6-3x^4)^{10}$

$$y' = 4(x^2+3x-1)^3(2x+3)(6-3x^4)^{10}$$
$$+ (x^2+3x-1)^4(10)(6-3x^4)^9(0-12x^3)$$

Ex. $y = \sqrt{x}\ \sqrt[4]{x^2+8x-9}$

$$y' = \frac{1}{2}x^{-\frac{1}{2}}\sqrt[4]{x^2+8x-9} + \sqrt{x}\left(\frac{1}{4}\right)(x^2+8x-9)^{-3/4}(2x+8)$$

Ex. $y = \left(\sqrt{x} - \frac{1}{\sqrt{x}}\right)\left(\frac{10}{x^2} - 6x\right)$

$$y' = \left(\frac{1}{2}x^{-\frac{1}{2}} - \left(-\frac{1}{2}\right)x^{-3/2}\right)\left(\frac{10}{x^2} - 6x\right)$$
$$+ \left(\sqrt{x} - \frac{1}{\sqrt{x}}\right)\left(10(-2)x^{-3} - 6\right)$$

Rule:

$$\frac{d}{dx}\frac{f(x)}{g(x)} = \frac{f'(x)g(x) - g'(x)f(x)}{g^2(x)}$$

Ex: $y = \dfrac{x^3+x^2-1}{7x^2-8x+4}$

$$y' = \frac{(3x^2+2x)(7x^2-8x+4) - (14x-8)(x^3+x^2-1)}{(7x^2-8x+4)^2}$$

Ex: $y = \dfrac{x-\sqrt{x}}{x+\sqrt{x}}$

$$y' = \frac{(1-\frac{1}{2}x^{-\frac{1}{2}})(x+\sqrt{x}) - (1+\frac{1}{2}x^{-\frac{1}{2}})(x-\sqrt{x})}{(x+\sqrt{x})^2}$$

Ex: $y = \left(\dfrac{2x-3}{7x+4}\right)^6$

$$y' = 6\left(\frac{2x-3}{7x+4}\right)^5 \left[\frac{2(7x+4) - 7(2x-3)}{(7x+4)^2}\right]$$

Rule Summary:

$y(x)$	$y'(x)$
x^n	$n x^{n-1}$
x	1
c	0
cx^n	$cn x^{n-1}$
$c f(x)$	$c f'(x)$
$(f(x))^n$	$n(f(x))^{n-1} f'(x)$
$f(x)\, g(x)$	$f'(x) g(x) + f(x) g'(x)$
$\frac{f(x)}{g(x)}$	$\frac{f'(x) g(x) - g'(x) f(x)}{g^2(x)}$

Exercise Find $y'(x)$:

① $y = 6 - 3x^8 + \frac{1}{6x^2} - 7\sqrt{x} - 3x$ ② $y = \sqrt[3]{\frac{x^2-1}{6-8x}}$

③ $y = \sqrt{x^4 - 6x + 1}\,(2x-1)^8$ ④ $y = \frac{1}{\left(\frac{3}{x^4} - \frac{2}{\sqrt{x}}\right)}$

Solutions

① $y' = 0 - 24x^7 + \frac{1}{6}(-2)x^{-3} - 7\left(\frac{1}{2}\right)x^{-\frac{1}{2}} - 3$

② $y' = \frac{1}{3}\left(\frac{x^2-1}{6-8x}\right)^{-2/3}\left[\frac{2x(6-8x) - (-8)(x^2-1)}{(6-8x)^2}\right]$

③ $y' = \frac{1}{2}(x^4 - 6x + 1)^{-\frac{1}{2}}(4x^3 - 6)(2x-1)^8$

$+ \sqrt{x^4 - 6x + 1}\,(8)(2x-1)^7(2)$

④ $y = \left(\frac{3}{x^4} - \frac{2}{\sqrt{x}}\right)^{-1}$

$y' = (-1)\left(\frac{3}{x^4} - \frac{3}{\sqrt{x}}\right)^{-2}\left(3(-4)x^{-5} - 2\left(\frac{1}{2}\right)x^{-\frac{1}{2}}\right)$